Tutorial 4

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Power series

Problem 16 Show that the series $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ converges absolutely for all $z \in \mathbb{C}$ with $|z| \leq 1$, and diverges for all $z \in \mathbb{C}$ with $|z| > 1$. In this case, the domain of convergence is a closed disc.

Problem 17 Show that the series $\sum_{n=1}^{\infty} n z^n$ converges absolutely for all $z \in \mathbb{C}$ with $|z| < 1$, and diverges for all $z \in \mathbb{C}$ with $|z| \geq 1$. In this case, the domain of convergence is an open disc.

Problem 18 Determine the radii of convergence of the following power series:

(a) $\displaystyle\sum_{n=0}^{\infty} (-1)^n (z-i)^n$

(b) $\displaystyle\sum_{n=0}^{\infty} (-1)^n 2^n z^{2n+2}$

(c) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n^2}{(2n+1)(n+2)}\right)^n z^n,$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{z^n}{\sqrt{n}},$

(e) $\displaystyle\sum_{n=0}^{\infty} \frac{n!}{n^n} (z+2)^n,$

(f) $\displaystyle\sum_{n=0}^{\infty} 2^n z^{(2^n)}.$

Problem 19 Use the geometric series to determine the power series expansion of the form $\sum_{n=0}^{\infty} a_n z^n$ for the following functions:

(a) $\dfrac{1}{2z+5},$ (b) $\dfrac{1}{1+z^4},$ (c) $\dfrac{1+iz}{1-iz},$ (d) $\dfrac{1}{(z+1)(z+2)}.$

For each series, give its domain of convergence.